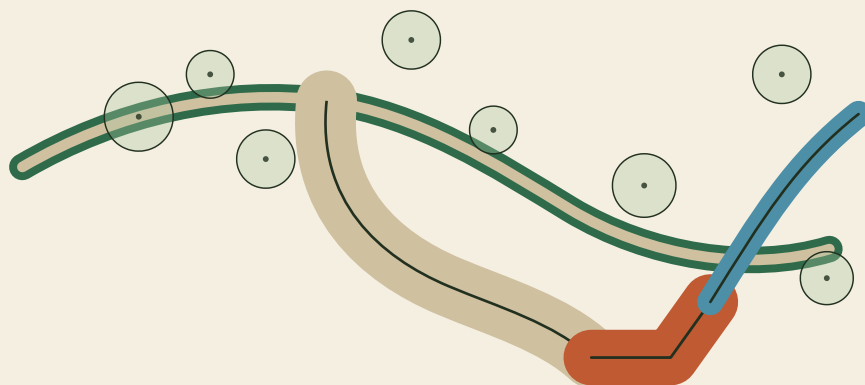




nabta  WELLNESS PARK

THE LIVING SHADE LOOP

DUBAI MUNICIPALITY • AI PARK DESIGN CHALLENGE • DRAFT COMPETITION ENTRY

D-06

Concept & Preliminary Masterplan

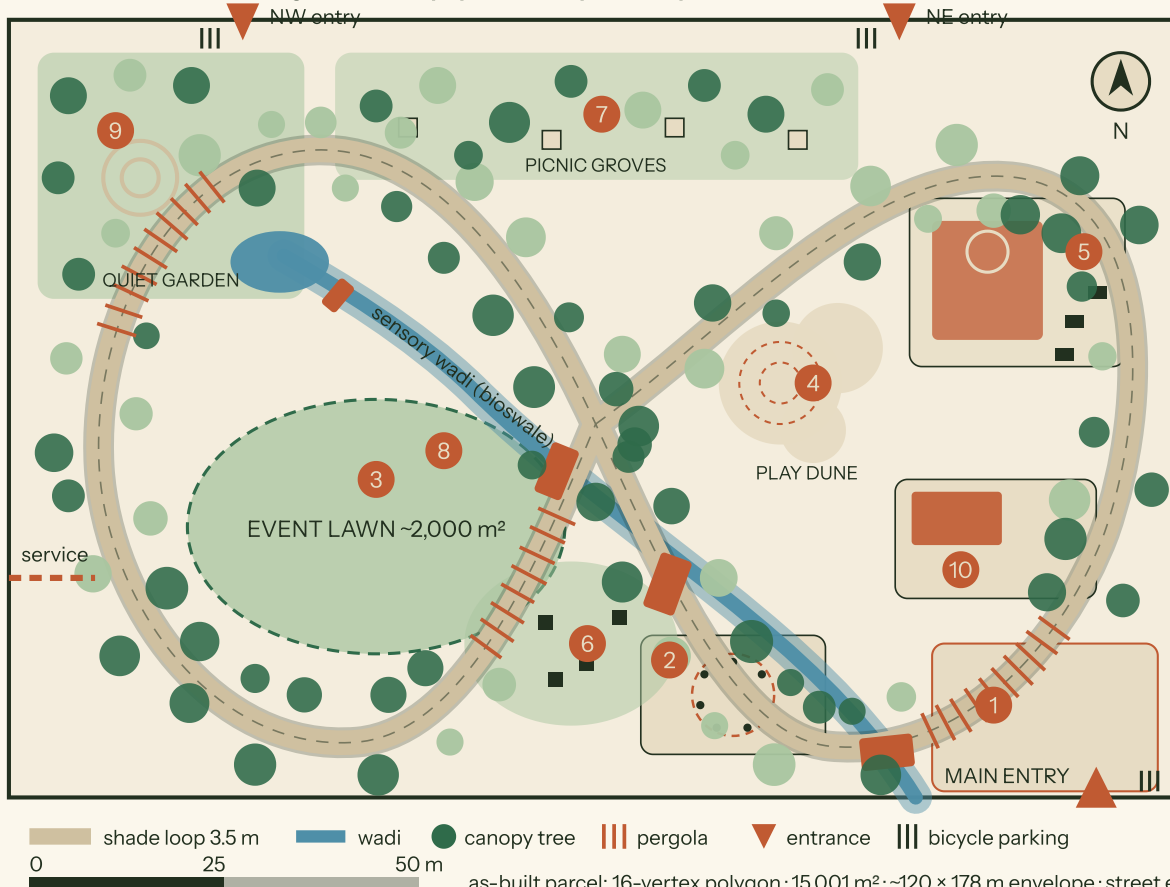
06 Masterplan



Deliverable 6 — Concept & Preliminary Masterplan

«نبطة» NABTA — Wellness Park: The Living Shade Loop. The plan sits on the true as-built parcel — an irregular 16-vertex polygon of 15,001 m² within a ~120 × 178 m envelope, street edge to the south, internal grid rotated ~47° from north — and is organized by one move: a continuous 3.5 m shaded path, a 398 m figure-eight braided into a 460 m perimeter ring for an 858 m continuous network (~1 km including the wadi spur). The design team selected the figure-eight through an open generative search: thousands of shade-element configurations scored in simulation (published in the runnable nabta-loop-optimizer repository) and four full concept variants scored against the site's real sun and wind (nabta-earth-obs — variant C won at 0.72); the leading options were stress-tested against six personas and an 1,800-visitor crowd simulation before this human-led choice, and community workshops are planned to validate it before design freeze. The loop is the park's spine and its climate device — under a canopy of ~450 trees for most of its length, under AI-form-found pergola segments where trees cannot mature quickly. It threads ten garden rooms arranged as a gradient: *active* rooms (entry plaza, majlis, café, play, teen zone) cluster at the south-east main entrance; *calm* rooms (picnic groves, quiet garden) retreat to the north and north-west. A sensory wadi — a planted bioswale — cuts the diagonal from the entry side to a still pool in the quiet garden, crossed by terracotta bridges where the loop passes over it.

«نبطة» NABTA — The Living Shade Loop · preliminary masterplan



PRELIMINARY MASTERPLAN, DIAGRAMMATIC — TO BE REDRAWN ON THE AS-BUILT CAD BASE (IRREGULAR 16-VERTEX PARCEL, GRID ~47° OFF NORTH). FIGURE-EIGHT 398 M + PERIMETER RING 460 M = 858 M CONTINUOUS SHADED NETWORK; ~1 KM INCLUDING THE WADI SPUR. ROOMS 1–10 KEYED IN THE SCHEDULE BELOW.

The crossing at the centre of the figure-eight is the plan's social hinge: every circuit passes it, so the fitness grove, event lawn edge and wadi bridges share one naturally surveilled node. The network is sized for ~1,800 daily visits. Room areas below are net; the path network is counted as circulation, and buffer planting carries the acoustic berm along the road edge. All figures are preliminary and will be tested against the AED 35M cost plan in Deliverable 8.

AREA SCHEDULE

#	GARDEN ROOM	AREA (M²)	% OF SITE
1	Entry plaza & gateway (SE)	900	6.0
2	Community majlis plaza	700	4.7
3	Event lawn (flexible)	2,000	13.3
4	Inclusive play dune	950	6.3
5	Teen active zone	850	5.7
6	Fitness grove	750	5.0
7	Picnic groves (N edge)	1,300	8.7
8	Sensory wadi garden	1,100	7.3
9	Quiet contemplation garden (NW)	900	6.0
10	Café terrace + restrooms	550	3.7
—	Shade network (3.5 m × 858 m)	3,000	20.0
—	Buffer planting, berms & service	2,001	13.3
Total		15,001	100.0

ZONING NOTES

- **Active-to-calm gradient.** Noise-generating rooms (1, 2, 4, 5, 10) sit within 60 m of the SE entry; rooms 7 and 9 occupy the northern band, buffered by the event lawn and wadi in between. gradient SE→NW
- **30 m road buffer.** No seating or play surface within 30 m of the southern main road except the hardscaped entry plaza; a planted berm and the densest tree massing line the S and E edges to cut traffic noise by an estimated 6–8 dB(A).
- **Prayer-time adjacency.** The majlis plaza (2) and café terrace (10) flank the main entry on the direct walking line to the neighbourhood mosque, supporting pre- and post-maghrib gathering; the quiet garden (9) offers a low-stimulus alternative.
- **School-run adjacency.** Play dune (4) and teen zone (5) cluster at the NE secondary entrance on the school walking route, absorbing the 14:00–16:00 peak without crossing calm rooms; bicycle parking at all three entrances.
- **Canopy budget.** ~450 trees (60% native ghaf, sidr and date palm; 40% proven adaptive shade species) are placed by a shade-simulation model and checked by the landscape team: target 92% of the 858 m network shaded at 15:00 in September (simulation verifies 91.6%), with pergolas bridging early-year gaps until year-5 canopy closure; the mature canopy cuts perceived temperature by up to 4.5 °C. Smart irrigation on recycled water cuts potable demand by 41% — a deliberately conservative figure. AI-assisted, human-verified

- **Service without conflict.** Maintenance access enters from the W edge, reaching the lawn and wadi directly and never crossing the loop at grade during operating hours.

IMPLEMENTATION PHASING – 18 MONTHS, PARK NEVER FULLY CLOSED

